



Chapter	Title	Total periods	Total time	Date
Ch 07	How Things Work	8	300 min	30 May 2026

C No.	Targeted competencies
C-1.3	Asks questions and makes predictions about simple patterns (season change, food chain, phases of the moon, movement of stars and planets, shapes of trees, plants, leaves, and flowers, rituals, celebrations) observed in the immediate environment
C-1.5	Uses local materials to create simple objects (family tree, envelopes, origami animals) on their own for display or use in classroom processes
C-6.1	Performs simple inquiry related to specific questions independently or in groups
C-6.2	Presents observations and findings through different creative modes (drawing, diagram, poem, play, skit, oral and written expression)
C-7.1	Gains familiarity with using the scientific method in investigations, as well as familiarity with other crosscutting concepts such as energy, matter, and systems that apply across the domains of science and engineering
C-7.2	Gains familiarity with disciplinary core ideas in the natural sciences, as well as in engineering, technology, and applications of science, which reflect the content that will be learned across subject areas in later Grades

Period 1	35 min	Which Objects Spin? Observe and Record	Pedagogical approach: Observe and Record
----------	--------	--	---

Material	coin, bangle, pencil, stone, wooden top, eraser, worksheet/notebook
0-5	Teacher reads the opening story of Meera and Dhruv aloud; students share one object from daily life they have seen spinning.
5-10	Students individually list objects they think can spin and those they think cannot, then share briefly with a partner.
10-25	In pairs, students collect the six objects (coin, bangle, pencil, stone, wooden top, eraser), spin each one, and record observations in the textbook table - noting whether it spins well, briefly, or not at all.
25-32	Class discussion: Which objects spun well? What do those objects have in common? Students share observations aloud.
32-35	Teacher draws out the idea of shape and balance affecting spinning; students note one finding in their own words.

LO { *Students can observe and record whether physical objects behave consistently under the same action, and identify a pattern across multiple trials.*

Period 2	40 min	Build and Test Cardboard Spinners	Hands-on Investigation
----------	--------	-----------------------------------	------------------------

Material	cardboard pieces, toothpicks, empty ballpoint pen tube, scissors, ruler, coloured markers, notebook
0-5	Teacher shows the textbook spinner diagrams; students predict which spinner they think will spin best and why.
5-20	In groups of three, students construct at least two spinners - one with the toothpick through the centre and one with the toothpick off-centre - following textbook configurations.
20-30	Students test each spinner, observe how long and how steadily it spins, and record observations in a simple table in their notebooks.
30-37	Groups discuss: Which spinner did not spin properly? What could be the reason? Students refer to the discuss prompts on p. 108.
37-40	Teacher consolidates: the position of the axle relative to the centre of the shape determines spinning quality.

LO { *Students can test how changing a single variable in a constructed object affects its functional behaviour, and record observations systematically.*



Period 3	35 min	Predict, Test, and Discuss: Float or Sink?	Hands-on Investigation
Material	bucket or large basin with water, leaf, iron nail, empty katori (bowl), plastic bottle with lid, stone pieces, aluminium foil, notebook		
0-5	Teacher reads the Meera-Dhruv paper-boat scene; students recall a time they saw something float or sink unexpectedly.		
5-15	Students individually fill in their predictions (F/S) for each object in the textbook table before any testing begins.		
15-27	In groups, students drop each object into the water basin and record the actual result (F/S) beside their prediction.		
27-33	Class discussion: Did all heavy objects sink and all light objects float? Students name exceptions and try to explain them.		
33-35	Teacher poses the key question for next period: 'Can we make something float just by changing its shape?'		
LO	{ Students can make and compare predictions about physical outcomes against observed results, and identify cases where an expected pattern does not hold.		

Period 4	35 min	Shape Changes Everything: Aluminium Foil Experiment	Hands-on Investigation
Material	aluminium foil (three equal pieces per group), bucket or basin with water, notebook		
0-5	Teacher recaps previous finding: weight alone doesn't decide. Introduces question: 'What if we change only the shape but keep the material the same?'		
5-20	Groups take three identical pieces of aluminium foil, shape them as instructed (spread out flat, pressed into a tight ball, folded into a cup), drop each into water, and record F or S plus their reason in the table on p. 112.		
20-28	Groups share results; teacher facilitates comparison - all three pieces were the same material and same amount, yet different outcomes.		
28-33	Students write one sentence in their notebooks explaining why shape affects floating or sinking, using their own words.		
33-35	Teacher connects to boat design: 'What shape would you use if you wanted to build a boat that carries the most weight?'		
LO	{ Students can isolate shape as a variable and demonstrate how the same material produces different physical outcomes when its form changes.		

Period 5	40 min	Design and Build a Boat	Create and Express
Material	paper, cardboard, ice-cream sticks, clay, adhesive tape, scissors, pencils/crayons for drawing, bucket or basin with water		
0-7	Groups of 3-4 students discuss boat design using available materials; each student sketches their boat idea and the group agrees on one plan.		
7-27	Groups build their boats, ensuring their design differs from other groups'; teacher circulates and asks groups to explain their design choices.		
27-35	Each boat is tested in the water basin; groups observe and note strengths and challenges in the comparison table on p. 114.		
35-40	Brief gallery walk - groups look at others' boats and note one strength of another group's design before the class exhibition is formally opened.		
LO	{ Students can apply understanding of shape and material properties to design, build, and evaluate a functional object, and communicate their reasoning to others.		

Period 6	40 min	Comparing and Reflecting on Boat Designs	Discussion and Connection
Material	completed boats from Period 5, Textbook (Section: Floating and Sinking) comparison table (p. 114), notebook		



0-8	Each group presents their boat for 1-2 minutes: materials used, design choice, and one thing they would change if they built it again.
8-20	Students individually complete the textbook comparison table: strengths and challenges of their own boat versus at least one other group's boat.
20-32	Whole-class discussion: What design features made some boats stronger or more stable? Students connect back to the aluminium foil experiment - how shape matters.
32-38	Teacher facilitates a brief reflection: 'People across the world have made boats from what was around them - bark, reeds, shells. What materials near your home could you use?'
38-40	Students write one sentence: the most important thing they learned about floating and sinking from the boat challenge.
LO	<i>Students can compare the functional outcomes of different design choices, identify strengths and limitations, and connect material-and-shape reasoning to real-world contexts.</i>

Period 7	35 min	Framing Questions and Reasoning About Balance	Discussion and Connection
Material	Textbook (Section: Let us reflect) (p. 115), notebook, pencil		
0-6	Teacher reads the scenario of Ravi's spinning observation aloud; students think silently and write at least two questions Ravi could ask his teacher.		
6-14	Pairs share their questions; class lists the best questions on the board and discusses what kind of investigation could answer each one.		
14-26	Teacher shows the textbook mobile figure (p. 115); students examine it and write their answer to what should be done to balance the mobile bending towards side A.		
26-32	Class discussion: What does it mean for something to be balanced? How does this connect to the spinners from Period 2 - why does a centred axle help a spinner stay steady?		
32-35	Students note one connection between balance and spinning in their notebooks.		
LO	<i>Students can formulate testable inquiry questions about an observed phenomenon and apply reasoning about balance to explain a physical scenario.</i>		

Period 8	40 min	Classify, Reason, and Reflect on Float and Sink	Reflect and Act
Material	Textbook (Section: Let us reflect) (p. 115-116), notebook, pencil		
0-7	Students read the reflection question on p. 115: How would you make a floating object sink and a sinking object float? They write their individual answers before any discussion.		
7-18	Students classify the fourteen objects on p. 116 (wax, marble, thermocol, candle, coin, cork, leaf, eraser, spoon, ice-cube, potato, tomato, pumpkin, lemon) into floaters and sinkers using reasoning - not physical testing.		
18-28	Pairs compare their classifications and discuss any disagreements; teacher highlights contested items (ice-cube, lemon, potato) and asks: 'What makes you uncertain?'		
28-36	Whole-class consolidation: students revisit their earlier predictions from Period 3 and reflect - what do they now know that they did not know at the start of the chapter?		
36-40	Students write one sentence about what they would investigate next if they could design their own experiment, closing the chapter on an open, curious note.		
LO	<i>Students can apply criteria developed through investigation to classify unfamiliar objects, and articulate how their reasoning has changed as a result of inquiry.</i>		